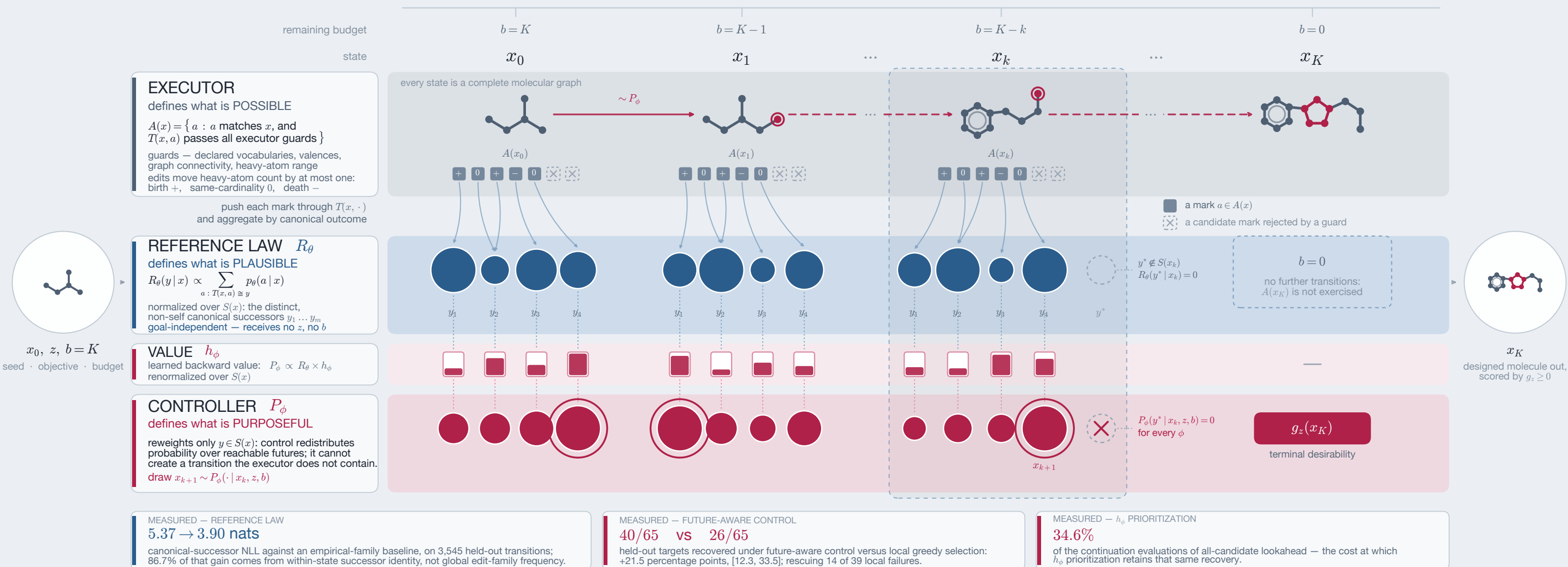


CONTROLLED KERNEL

$$P_{\phi}(y|x, z, b) = \frac{R_{\theta}(y|x) h_{\phi}(y, z, b-1)}{\sum_{y' \in S(x)} R_{\theta}(y'|x) h_{\phi}(y', z, b-1)}$$



TEMPORALLY EXTENDED ACTION

BUILD_RING_SYSTEM(topology, size, stoichiometry, state)

the controller learns factorized distributions over the four parameters:

topology $\in \{\text{linked, fused}\}$ **size**an integer over the
executor-qualified range**stoichiometry**an exact count vector over ring
elements, summing to size**state** $\in \{\text{aromatic, saturated}\}$ the compiler enumerates concrete realizations ξ (attachment site, heteroatom position), which collapse onto distinct canonical endpoints y_{ξ} :

$$P(\xi | a_{\text{semantic}}, x, z, b) \propto R_{\theta}(y_{\xi} | x) H(y_{\xi}, z, b)$$

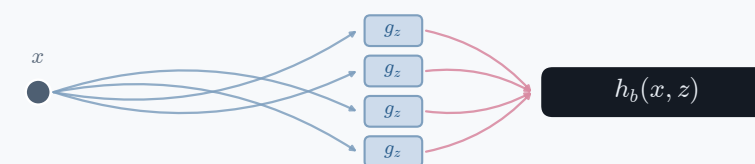
with $H = 1$ when no realization-level value model is availablerealizations ξ are NOT
controller actions

NOTATION

 x — a state: a complete molecular graph, never masked a — a typed rewrite mark; $T(x, a)$ — the graph after applying a $\cong$ — equivalence under canonicalization k — step index; K — total budget; $b = K - k$ — remaining budget $S(x)$ — distinct non-self canonical successors $y_1 \dots y_m$ of x y^* — a molecule outside $S(x)$: no legal mark produces it z — the design objective; $g_z \geq 0$ — terminal desirability ξ — a concrete realization; y_{ξ} — its canonical endpoint

BACKWARD VALUE

$$h_b(x, z) = \mathbb{E}_{R_{\theta}}[g_z(X_b) | X_0 = x]$$

 $g_z \geq 0$ — terminal desirability for objective z  b rollout steps under R_{θ} ; h_{ϕ} is a learned approximation to h_b at molecular scalewith exact backward values h_b , P_{ϕ} is the finite-horizon Doob h -transform of R_{θ} h_{ϕ} is a learned approximation at molecular scale; exact terminal reweighting is a property of the exact transform and is not claimed for h_{ϕ} .